

Financial Risk Analysis with Python

1. Project Overview

Financial institutions process thousands of transactions daily, making it essential to understand customer behavior and identify potential financial risks. This project analyzes Morgan Stanley customer transaction data to uncover transaction patterns, customer profiles, and risk indicators that can help improve customer service and financial monitoring.

The analysis was performed using Python and various data analysis libraries to clean, process, visualize, and interpret financial transaction data.

2. Objective

The main objectives of this project are:

- Perform data cleaning and preprocessing.
 - Analyze customer transaction behavior.
 - Build customer profiles based on activity levels.
 - Identify risky and suspicious financial activities.
 - Perform hypothesis testing on customer segments.
 - Generate meaningful visualizations and insights.
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3. Dataset Description

The dataset contains customer banking transaction records with information such as:

- Transaction ID
- Customer ID
- Account ID
- Account Type
- Transaction Type
- Product
- Region
- Transaction Date
- Transaction Amount
- Account Balance
- Risk Score
- Credit Rating
- Tenure Months

The dataset contains 800 records and 15 columns.

4. Technologies Used

- Python
 - Jupyter Notebook
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
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5. Data Cleaning and Preprocessing

The following preprocessing steps were performed:

- Loaded dataset using Pandas.
- Checked dataset dimensions and structure.
- Verified data types of all columns.
- Checked for missing values.
- Removed duplicate records.
- Converted transaction dates into datetime format.
- Verified data consistency after cleaning.

Result:

- No missing values found.
 - No duplicate records affecting analysis.
 - Date fields successfully converted.
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6. Exploratory Data Analysis (EDA)

Several analyses were performed to understand customer transaction behavior.

Transaction Analysis

Transaction amounts were analyzed across different transaction types:

- Deposit
- Withdrawal
- Transfer
- Payment

The analysis revealed transaction volume distribution and customer transaction preferences.

Account Type Analysis

Transactions were grouped according to account types to identify account usage trends.

Regional Analysis

Regional transaction behavior was examined to identify geographic patterns and differences in financial activities.

Customer Activity Analysis

Customer transaction frequency and account balances were analyzed to identify active and inactive customers.

7. Customer Segmentation

Customers were categorized into different groups based on transaction frequency and account balances.

High Activity Customers

Customers with frequent transactions and high account balances.

Medium Activity Customers

Customers showing moderate transaction activity and average balances.

Low Activity Customers

Customers with limited transaction activity and lower account balances.

8. Financial Risk Identification

Several risk assessment techniques were applied.

Risk Score Analysis

Customers with higher risk scores were identified and monitored.

Balance Volatility Analysis

Account balance fluctuations were analyzed to detect unstable financial behavior.

Anomaly Detection

Outliers and unusual transaction patterns were identified using statistical techniques.

Suspicious Activity Detection

Accounts showing irregular transaction behavior and large withdrawals were flagged for further review.

9. Hypothesis Testing

A statistical hypothesis test was conducted to determine whether customers with high transaction volumes maintain significantly higher account balances than customers with lower transaction volumes.

Hypothesis

H0: There is no significant difference in average account balances between high-volume and low-volume transaction customers.

H1: High-volume transaction customers have significantly higher average account balances.

The test results helped evaluate the relationship between transaction activity and customer balances.

10. Key Findings

- Customer transaction behavior varies significantly across account types.
 - High transaction activity often corresponds to higher account balances.
 - Certain customers exhibit elevated risk scores and unusual transaction patterns.
 - Regional differences influence transaction behavior.
 - Statistical analysis provides evidence for customer segmentation strategies.
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11. Business Recommendations

- Monitor high-risk accounts regularly.
 - Implement automated anomaly detection systems.
 - Develop targeted services for high-value customers.
 - Improve engagement strategies for low-activity customers.
 - Use transaction behavior insights for fraud prevention and risk management.
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12. Conclusion

This project successfully analyzed Morgan Stanley customer transaction data to identify financial behavior patterns, customer segments, and potential risks. The insights generated can help financial institutions improve decision-making, customer relationship management, and risk mitigation strategies.

The project demonstrates practical applications of Python, data analysis, visualization, customer segmentation, and statistical testing in the financial domain.